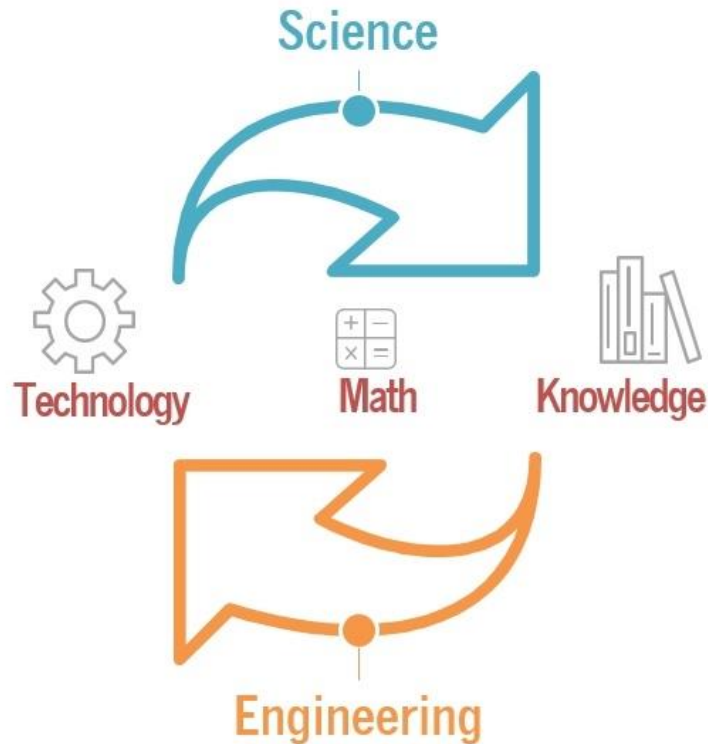


TTK4240 Industriell elektroteknikk



January 06, 2025

Marta Molinas
Institutt for teknisk kybernetikk

Introduction: The People



Teacher and main Lecturer:
Marta Molinas (marta.molinas@ntnu.no)

Overstudass:

Anker Ulrik Garmark	ulrikga@stud.ntnu.no

Stud. Ass.: 5 assistants



Vit.ass.:
Zeinab Karami (zeinab.k.h.abadi@ntnu.no)

Broch-Torp Phillip-André	philliab@stud.ntnu.no
Frisch Jørgen	jorgen.k.frisch@ntnu.no
Stensrud William	willijs@stud.ntnu.no
Warø Fredrik Sollie	fredrswa@stud.ntnu.no

Syllabus

Texts

1. Electric Circuits, Nilsson & Riedel
 2. Electrical Engineering Principles and Applications, Hambley
 3. Power Electronics, Mohan.
- PDFs are available on Blackboard
 - We recommend to study them
 - Not all chapters are relevant. See the lecture plan

Exercises and Laboratories

- **Øvinger:** Øving 1 (W2) - Øving 12 (W15)
 - Wednesdays from 14:15 to 16:00 (A1)
 - Total 12 with a minimum of 8 (deadline to be fixed by vit.ass.)
 - Contact: Zeinab Karami and overstudass.
- **Labs:** Digital Motor Lab (W4 to W15)
 - Mondays, Wednesdays, Thursdays and Fridays
 - <https://www.ntnu.edu/studies/courses/TTK4240#tab=timeplan>
 - Groups to be defined with vit.ass. via online registration
 - Contact Zeinab Karami, overstudass and studass.
- **Blackboard:** course information, lecture plan, folders for extra material (exercise plan, lab plan, videos, other info)

Exercises and Laboratories

12 assignments:

divided into 3 blocks, 8 of 12 required

Recommendation:

At least 2 of 4 assignments in each block.

Digital motor lab: compulsory.

Groups: 2-4 students, 4 lab-sessions, 3 hours each.

Start-End: week 3-week 15.

Signup deadline: TBD.

Exercise session



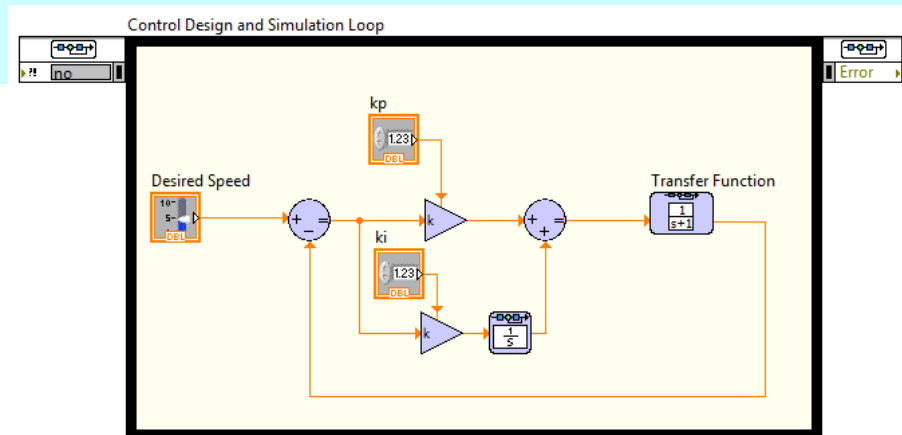
Lab session



Laboratory session

- Digital motorlab
 - Control of motor using Labview (graphic programming)
 - Max. 3 hours per session
 - Modeling of motor-load system
 - Design of control system

Labview



DC motor set



Reference group: for group feedback



- At least 3 persons from different study lines.
- Meetings with teacher, lab assistants and vit.ass 2 times during the semester.
- Send mail to zeinab.k.h.abadi@ntnu.no or contact on Collaborate

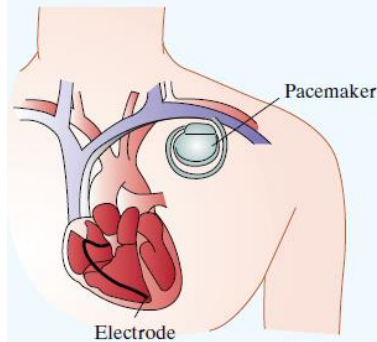
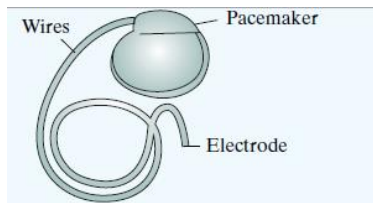
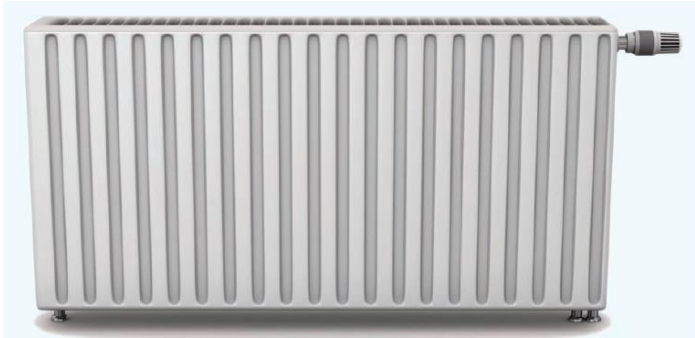
PART 1

Learning Objectives: Part 1

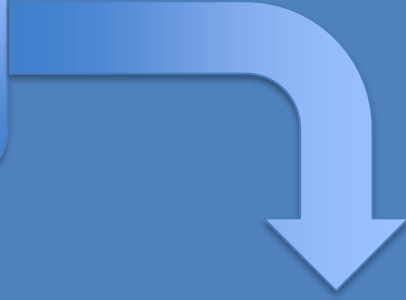
- **Basic Circuit Analysis**

- Frequency dependent circuits: passive filters
- Use of Laplace transform and transfer functions when designing controllers for industrial systems
- Electromagnetism and magnetic coupling
- Ideal transformer
- Stationary analysis of circuits with sinusoidal signals: current, voltage and power calculations

Applications: Low Power



Part 1

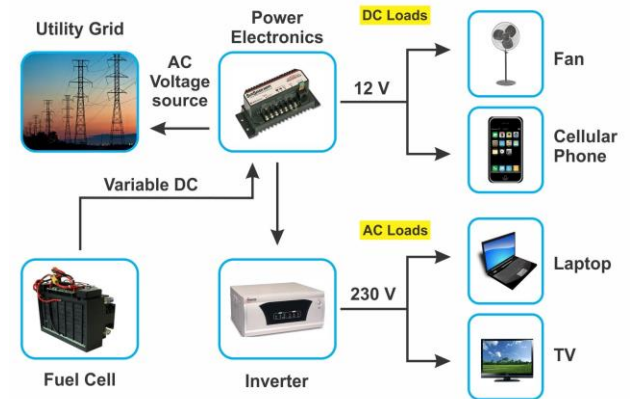
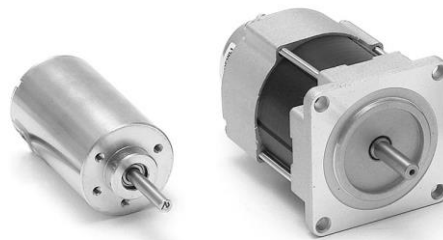
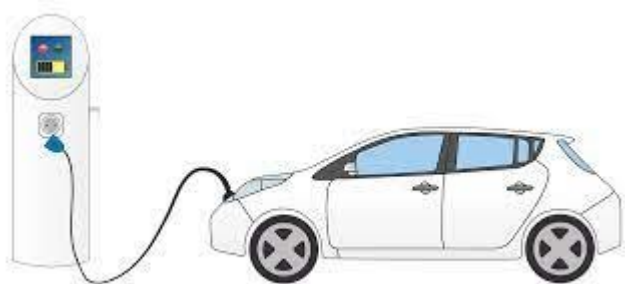
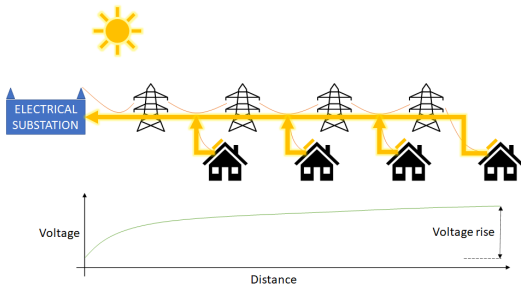


PART 2

Learning Objectives: Part 2 (from week 8)

- Electromechanical Energy Conversion:
 - DC and AC Motors
- Static Energy Conversion:
 - Static Rectifiers and Power supplies,
 - DC-to-DC Conversion,
 - Power electronics converters (AC-to-DC and DC-to-AC),
- Electricity Production, Distribution and Consumption:
 - Renewable and Non-renewable energy (Hydro, Wind, PV, etc)
 - Smart grid and microgrids
 - Energy Storage
 - Electrical Drive Systems (e.g. Electric Vehicle)

Applications: High Power



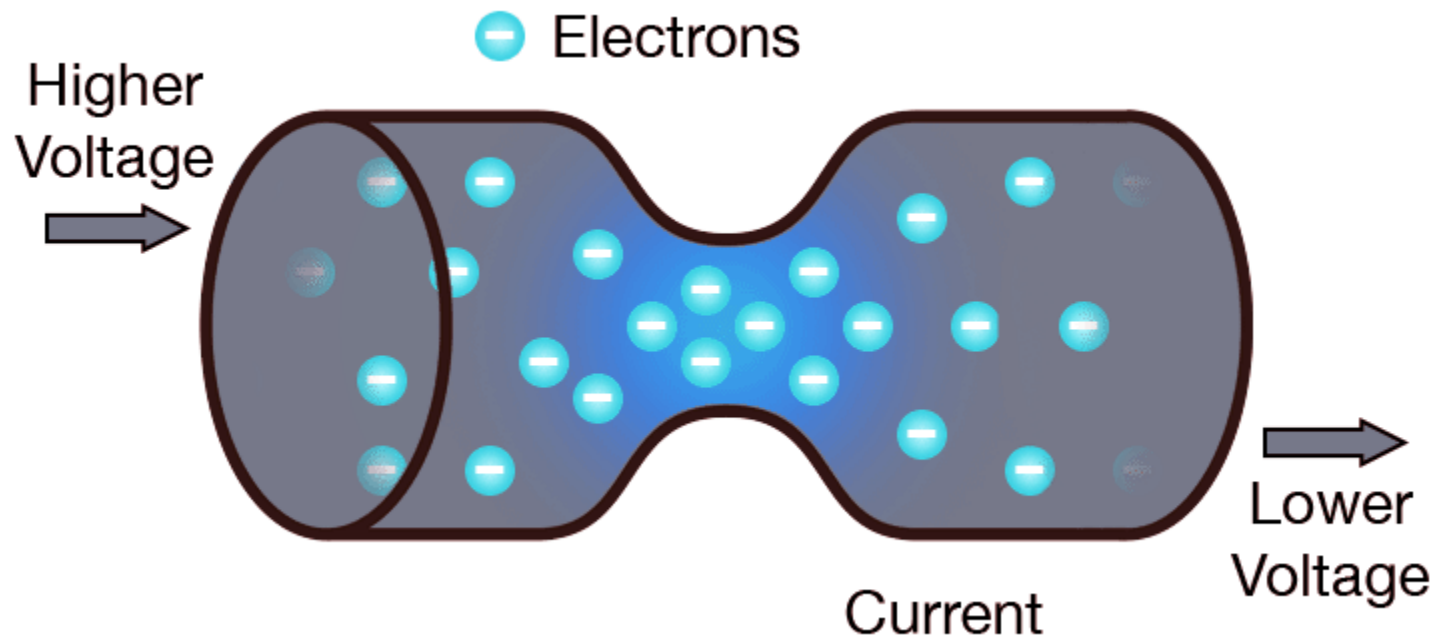
Lecture - Week 02 Part I

Review

- Voltage and Current: **Circuit Laws**
- Power and Energy
- **The 5 ideal basic «lumped» circuit elements**
 - Voltage and current sources
 - The Resistor
 - The Inductor
 - The Capacitor
- Natural and step response of RL circuit

Nilsson-Riedel: 1.4-1.6, 2.1, 2.2, 2.4, 3.1, 3.2, 4.5, 4.10, 4.12, 6.1-6.3, 7.1, 7.3

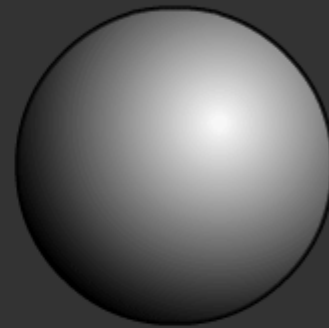
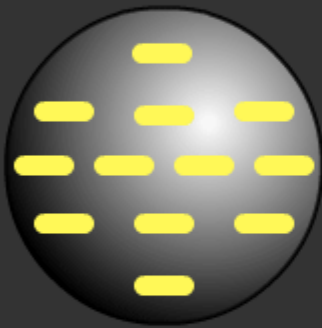
Voltage is the push that makes current to flow



https://www.youtube.com/watch?v=OGa_b26eK2c&t=472s

Voltage, current and resistance

Voltage (Potential Difference)



"Electric Pressure" Per Charge

Voltage and Current

$$v = \frac{dw}{dq},$$

v [volts]

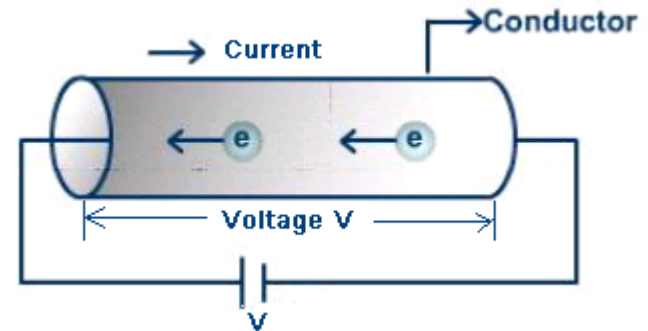
i [amperes]

q [coulombs]

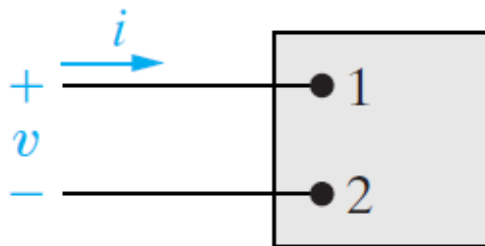
t [seconds]

w [joules]

$$i = \frac{dq}{dt},$$



- Electric pressure or energy (dw) per charge (dq) is **Voltage**
- Rate of charge (dq) flow is **Electric Current**

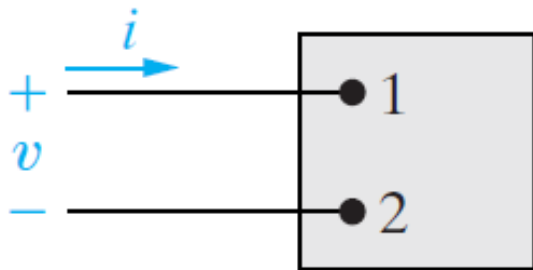


Power and Energy

$$p = \frac{dw}{dt},$$

$$= (dw/dq) \times (dq/dt) \\ = vi$$

$$p = vi$$



v [volts]

i [amperes]

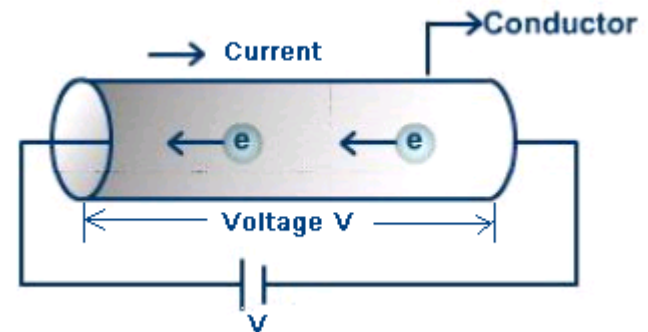
q [coulombs]

t [seconds]

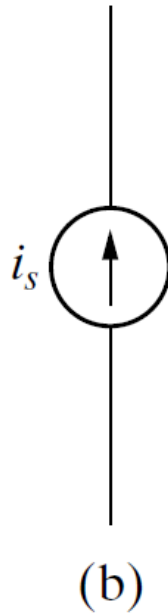
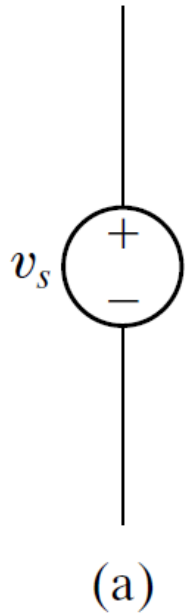
p [watts]

w [joules]

Change of energy
per unit time is
Power

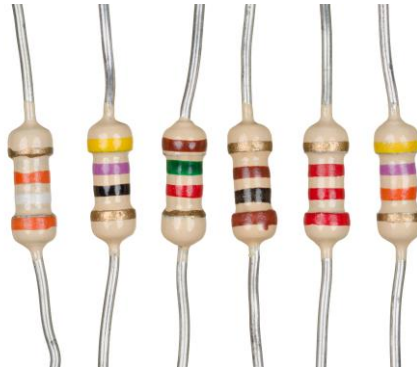


Ideal Voltage and Current Sources



- Independent voltage and current sources
(Active elements)

Resistor

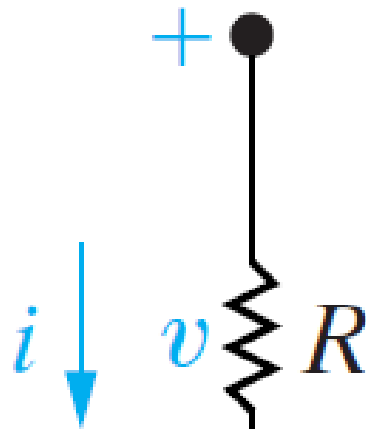


Electrical Resistance

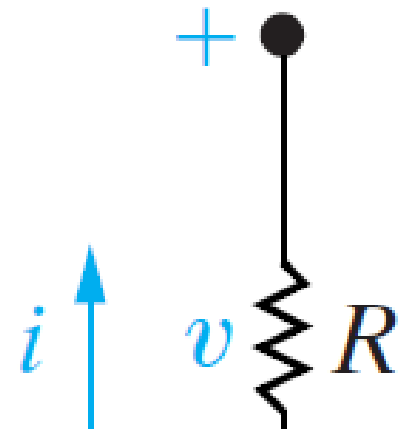
Ohm's Law

$$v = iR,$$

R: intrinsic property of a material

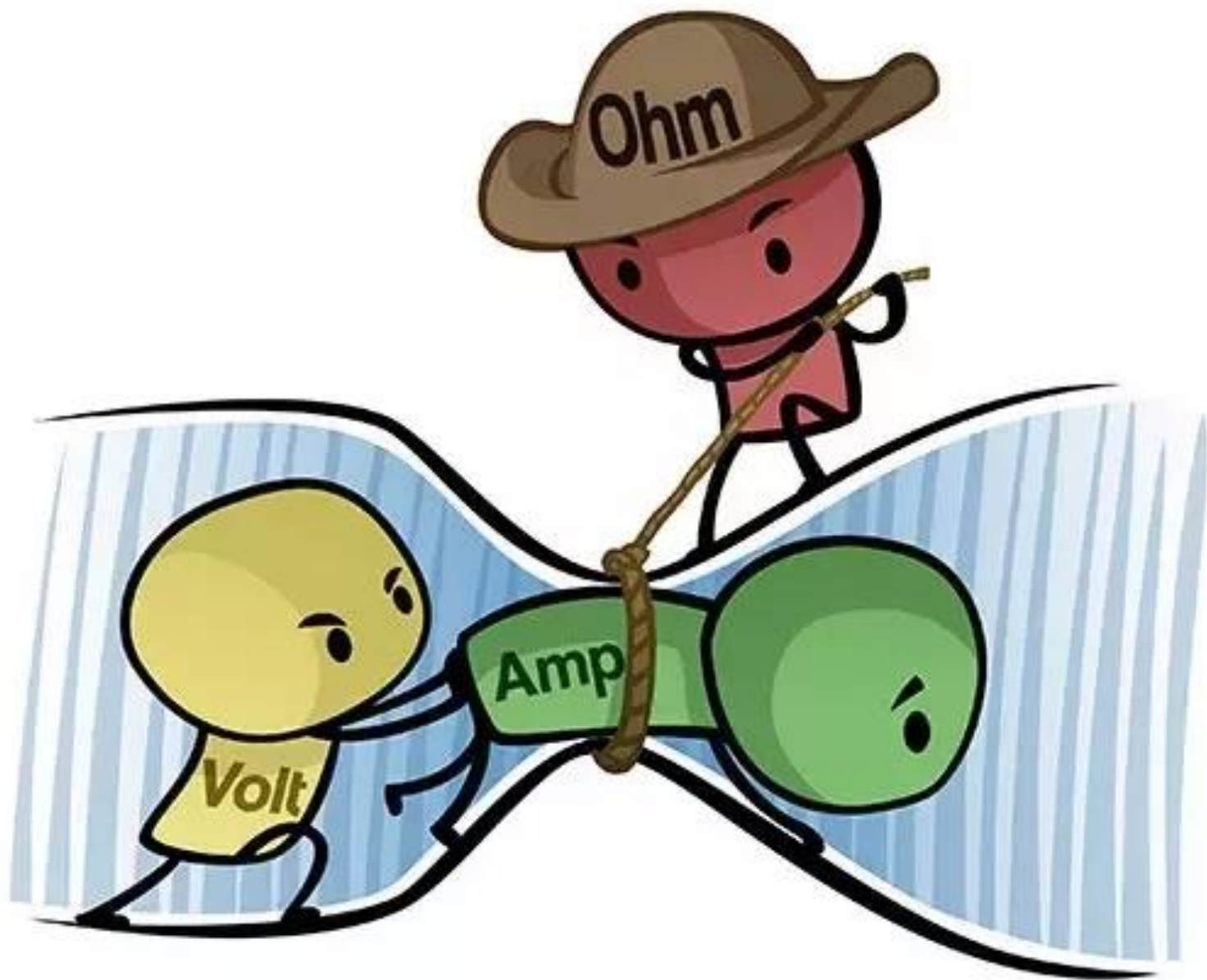


$$v = iR$$

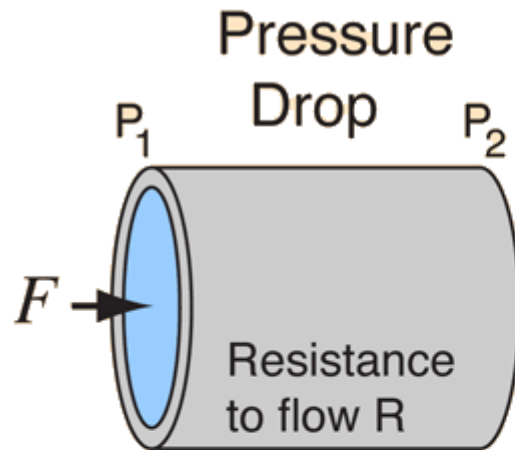


$$v = -iR$$

From the observation of phenomena to Physics laws

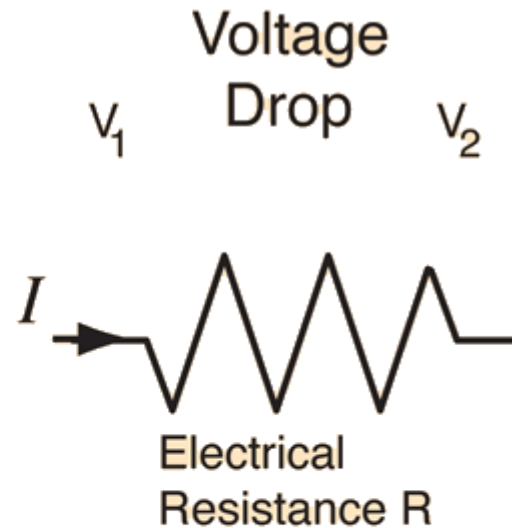


Resistor analogy



$$F = \frac{P_1 - P_2}{R}$$

*Poiseuille's law
for fluids*



$$I = \frac{V_1 - V_2}{R}$$

*Ohm's law
for electric circuits*

Power in a Resistor

$$p = i^2 R.$$

$$p = \frac{v^2}{R}.$$

v [volts]

i [amperes]

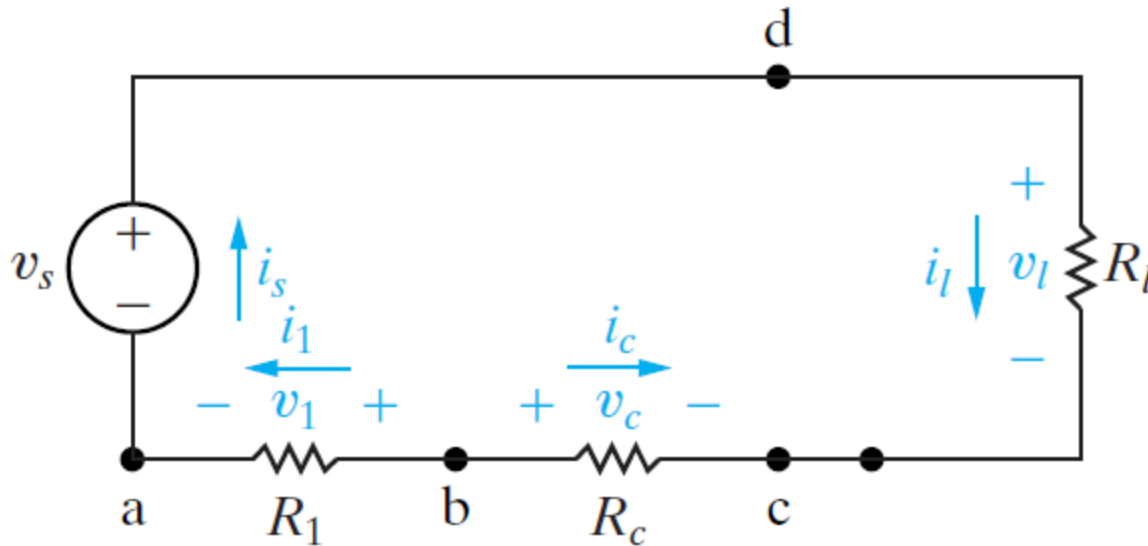
r [ohms]

Handwritten red derivations of power formulas for a resistor:

- Top: $P = v \times i$
- Middle: $v = i \times R$
- Bottom: $P = i^2 \times R$

Arrows indicate the substitution of $v = i \times R$ into $P = v \times i$ to derive $P = i^2 \times R$.

Kirchhoff's Current Law (KCL)



node a $i_s - i_1 = 0,$

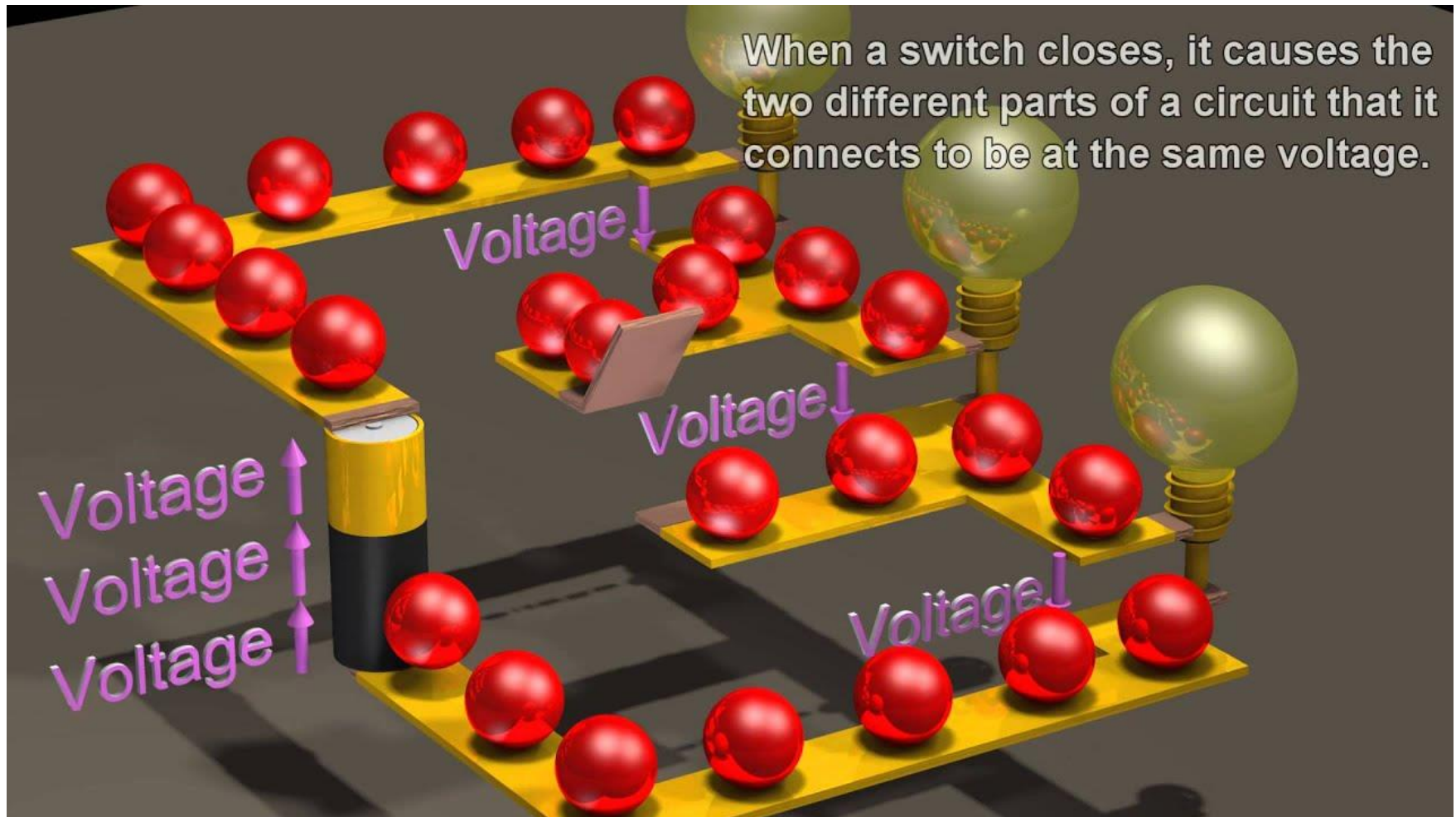
node b $i_1 + i_c = 0,$

node c $-i_c - i_l = 0,$

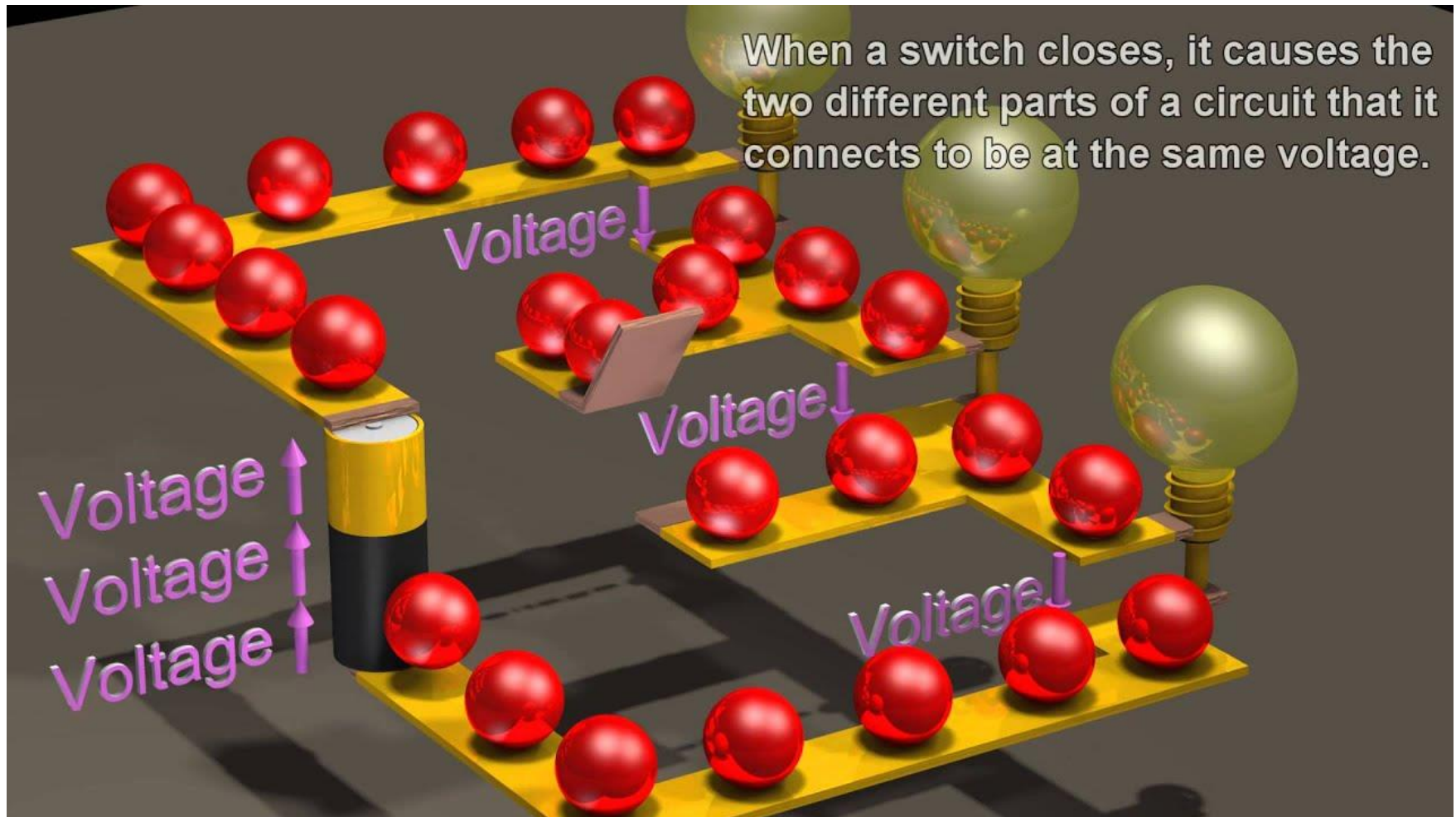
node d $i_l - i_s = 0.$

The algebraic sum of all the currents at any node in a circuit equals zero.

Kirchohoff' Voltage Law (KVL)

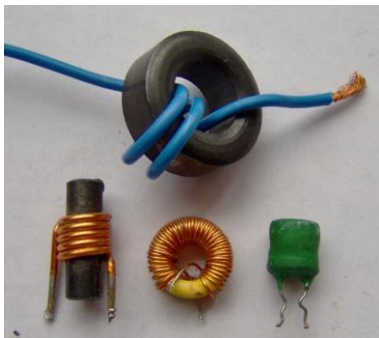


Basics of voltage and currents: video summary

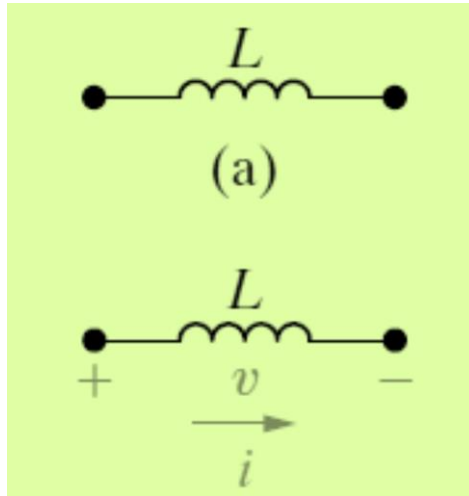


Break

Inductor & Capacitor



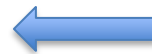
The Inductor v - i equation



$$v = L \frac{di}{dt},$$



$$i(t) = \frac{1}{L} \int_{t_0}^t v d\tau + i(t_0),$$



$$v dt = L di.$$

The current cannot change instantly in an inductor
Short circuit for constant or dc current ($di/dt=0$)

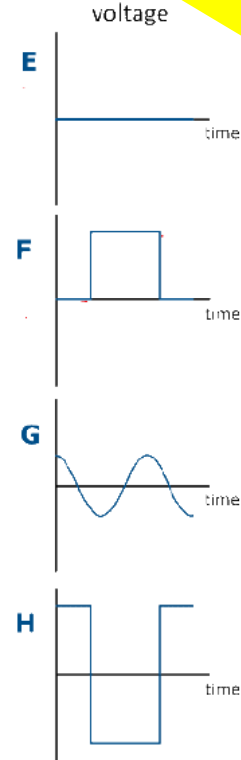
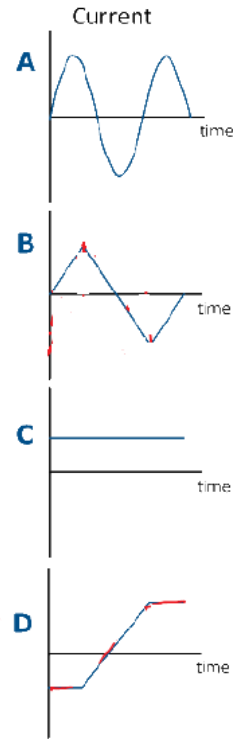
Key points

Q1: Match the current and voltage waveforms

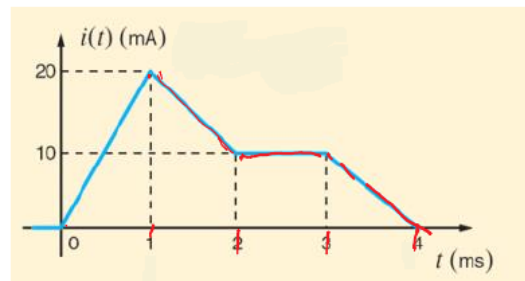
Q2: $L=5\text{mH}$, sketch $v(t)$

$$v = L \frac{di}{dt}$$

Q1



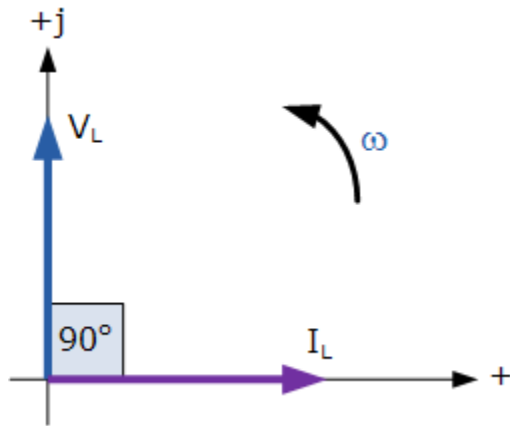
Q2



Current in an Inductor



$$v \, dt = L \, di.$$



$$i(t) = \frac{1}{L} \int_{t_0}^t v \, d\tau + i(t_0),$$

Power & Energy in the Inductor

$$p = vi.$$

$$p = Li \frac{di}{dt}.$$

$$p = \frac{dw}{dt} = Li \frac{di}{dt} \quad \rightarrow \quad dw = Li di.$$

$$\int_0^w dx = L \int_0^i y dy,$$

Stores electrical
energy in a
magnetic field.

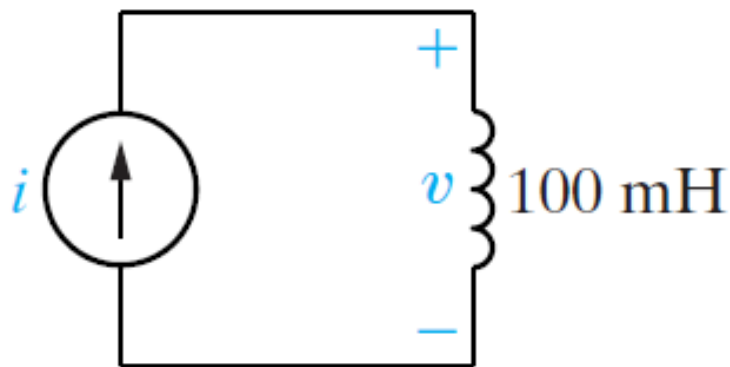
$$w = \frac{1}{2} Li^2.$$

Solve example 6.1

Exercise 6.3 (p.180 Nilson)

Exam type

- Determine voltage, current, power and energy of the inductor in the following circuit:



$$i = 0, \quad t < 0$$

$$i = 10te^{-5t} \text{ A}, \quad t > 0$$

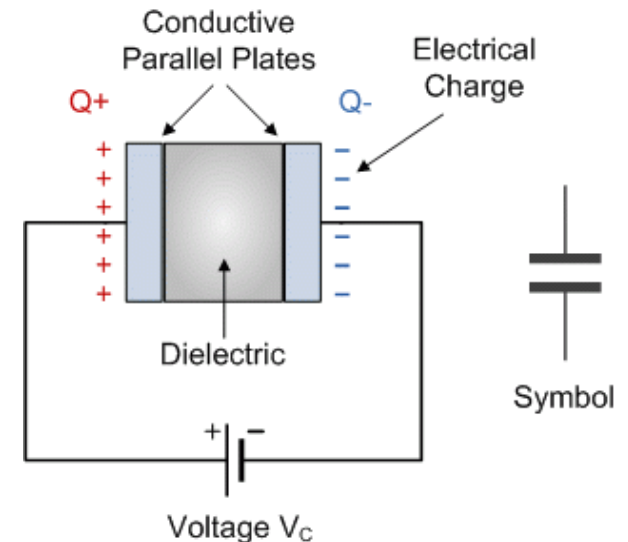
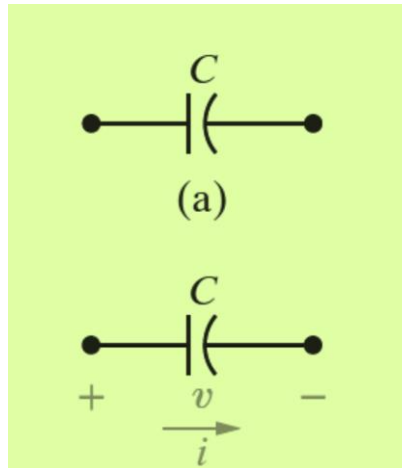
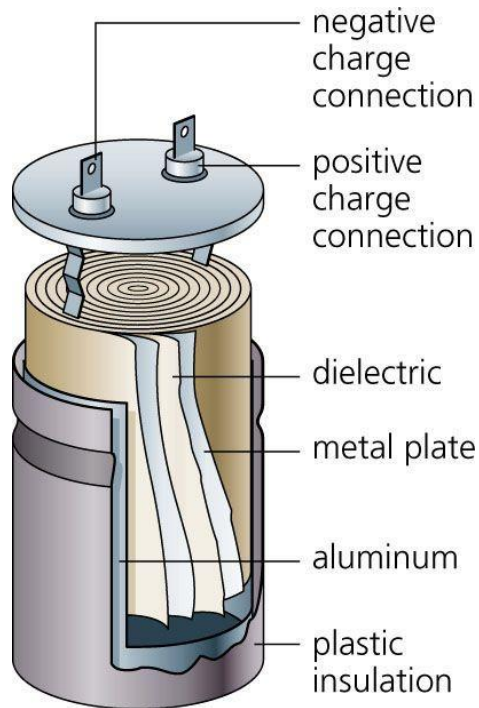
Check points

1. How much energy is stored in an inductor of 20mH with a current of 80mA?
2. What is the voltage on an inductor if $di/dt=0$ A/s? Is it a short or an open circuit situation?
3. Can the current through an inductor change instantaneously?
4. Will an inductor dissipate energy?



$$v = L \frac{di}{dt},$$

The Capacitor v - i equation

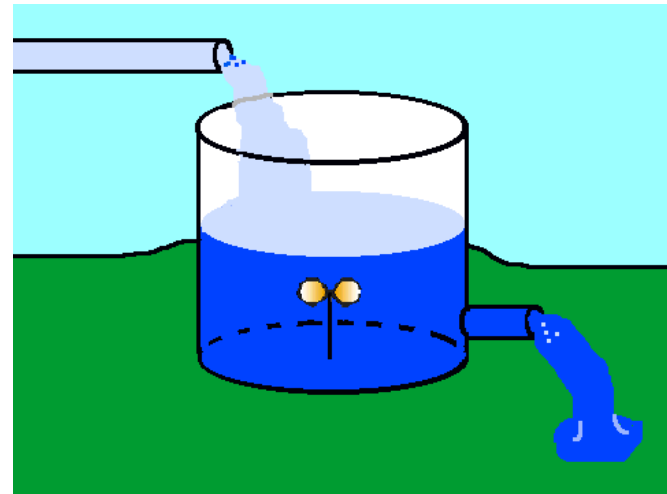
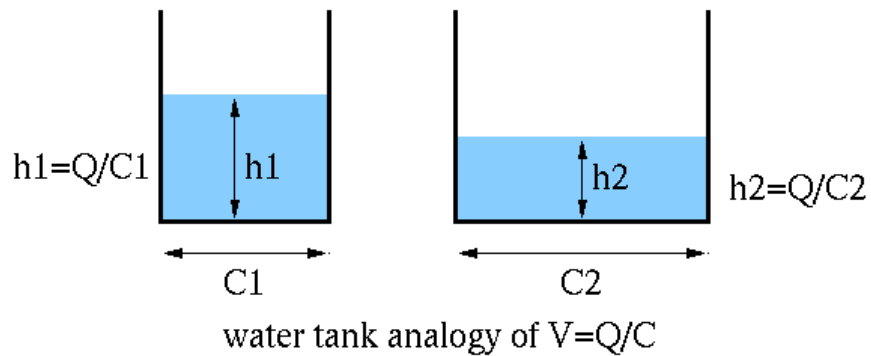


$$i = C \frac{dv}{dt},$$

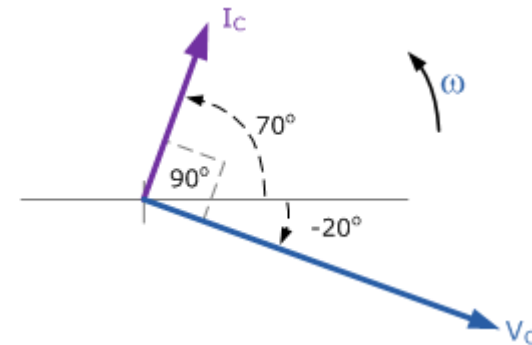
$$i dt = C dv \quad \text{or} \quad \int_{v(t_0)}^{v(t)} dx = \frac{1}{C} \int_{t_0}^t i d\tau.$$

$$v(t) = \frac{1}{C} \int_{t_0}^t i d\tau + v(t_0).$$

The Capacitor analogy



$$v(t) = \frac{1}{C} \int_{t_0}^t i \, d\tau + v(t_0).$$



https://wn.com/capacitor_analogy

Power & Energy in the capacitor

$$i = C \frac{dv}{dt},$$

$$p = vi = Cv \frac{dv}{dt},$$

$$dw = Cv dv,$$

$$\int_0^w dx = C \int_0^v y dy,$$



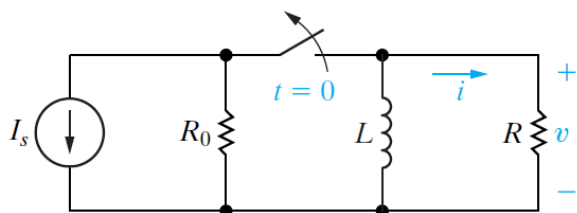
$$w = \frac{1}{2} C v^2.$$



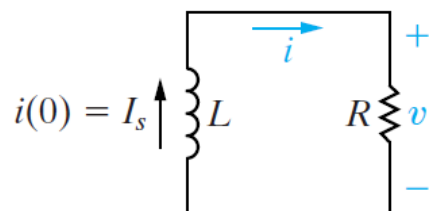
Stores electrical energy in an electric field.

Break

Natural response of a circuit



$$\frac{di}{dt}dt = -\frac{R}{L}i dt.$$



$$L \frac{di}{dt} + Ri = 0,$$

Natural response of a circuit

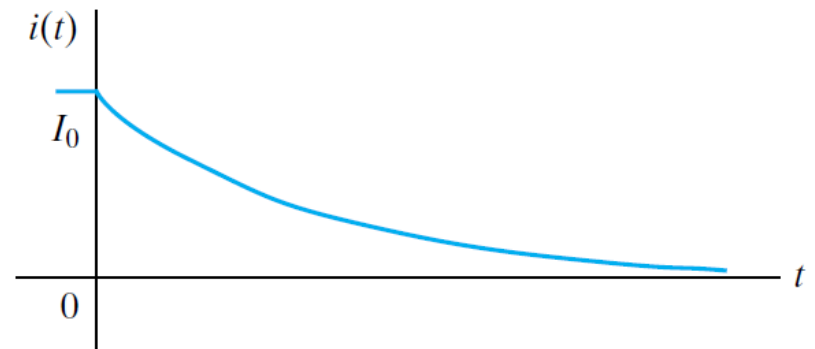
Natural response of RL circuit

$$\frac{di}{i} = -\frac{R}{L}dt. \quad \longrightarrow \quad \int_{i(t_0)}^{i(t)} \frac{dx}{x} = -\frac{R}{L} \int_{t_0}^t dy,$$

$$\ln \frac{i(t)}{i(0)} = -\frac{R}{L}t. \quad \longrightarrow \quad i(t) = i(0)e^{-(R/L)t}.$$

$$i(0^-) = i(0^+) = I_0,$$

$$i(t) = I_0 e^{-(R/L)t}, \quad t \geq 0,$$



Power and Energy in RL circuit

$$p = vi, \quad p = i^2 R, \quad \text{or} \quad p = \frac{v^2}{R}.$$

$$\begin{aligned} w &= \int_0^t p dx = \int_0^t I_0^2 R e^{-2(R/L)x} dx \\ &= \frac{1}{2(R/L)} I_0^2 R (1 - e^{-2(R/L)t}) \\ &= \frac{1}{2} L I_0^2 (1 - e^{-2(R/L)t}), \quad t \geq 0. \end{aligned}$$

Section 7.2: Natural response of an RC circuit

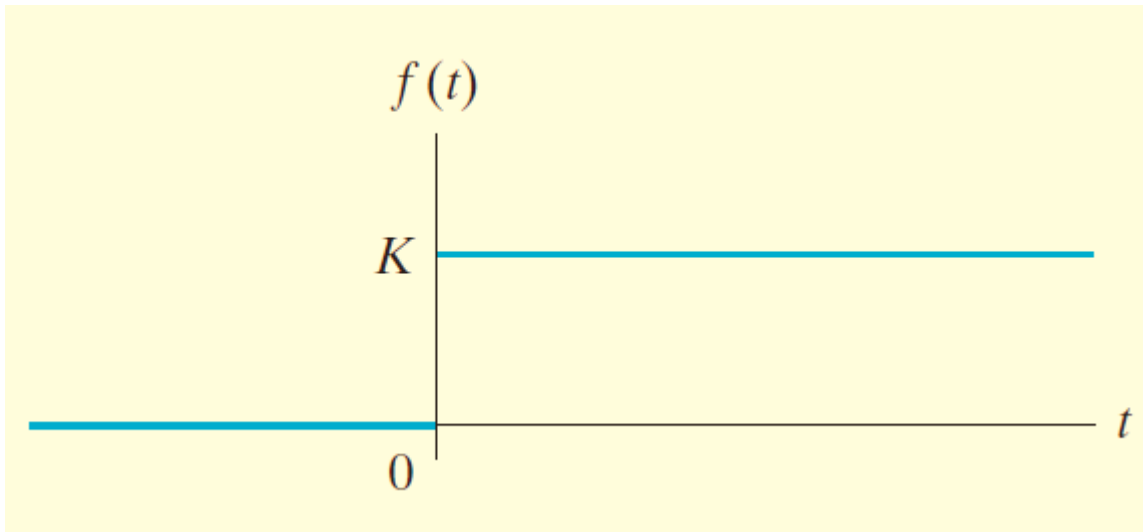
Homework

- Homework

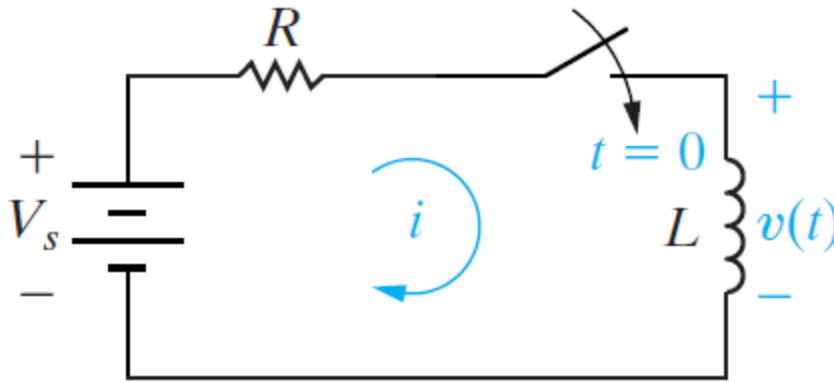
Step response of a circuit

Step response

Voltage and current signals when a constant source is applied to the circuit at a given time point.



Step response of RL circuit



$$V_s = Ri + L \frac{di}{dt},$$

$$\frac{di}{dt} = \frac{-Ri + V_s}{L} = -\frac{R}{L} \left(i - \frac{V_s}{R} \right).$$

Step response of RL circuit

$$di = \frac{-R}{L} \left(i - \frac{V_s}{R} \right) dt.$$

$$\frac{di}{i - (V_s/R)} = \frac{-R}{L} dt,$$

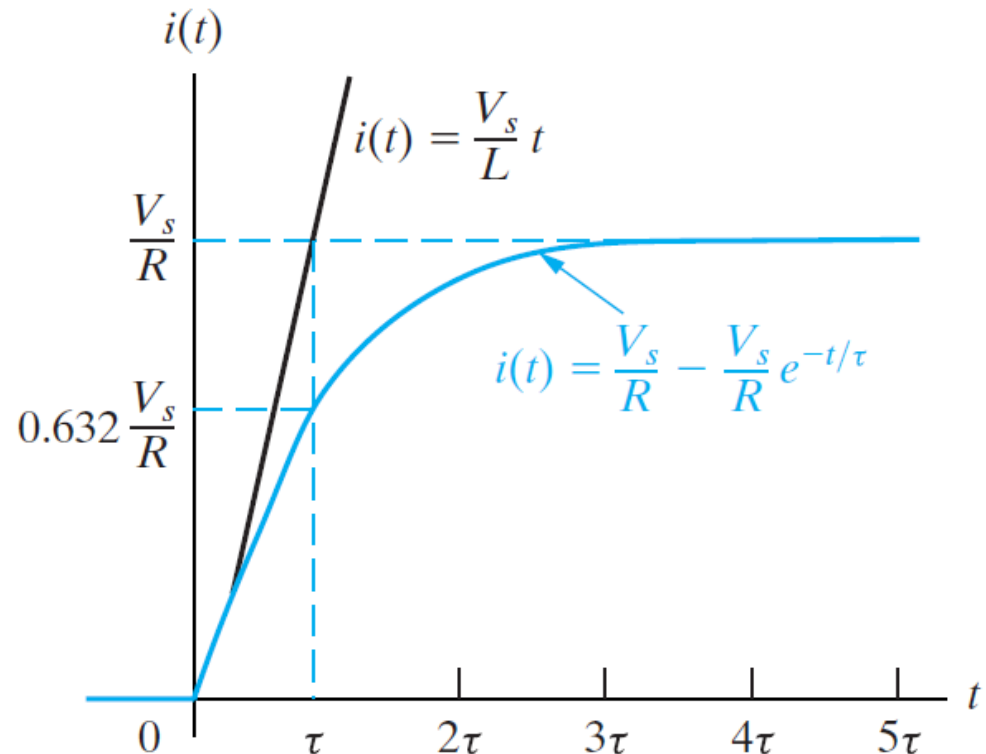
$$\int_{I_0}^{i(t)} \frac{dx}{x - (V_s/R)} = \frac{-R}{L} \int_0^t dy,$$

Step response of RL circuit

$$\ln \frac{i(t) - (V_s/R)}{I_0 - (V_s/R)} = -\frac{R}{L}t,$$

$$\frac{i(t) - (V_s/R)}{I_0 - (V_s/R)} = e^{-(R/L)t},$$

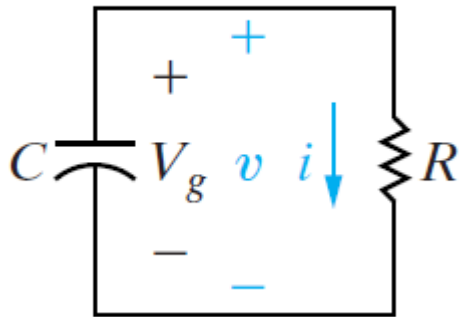
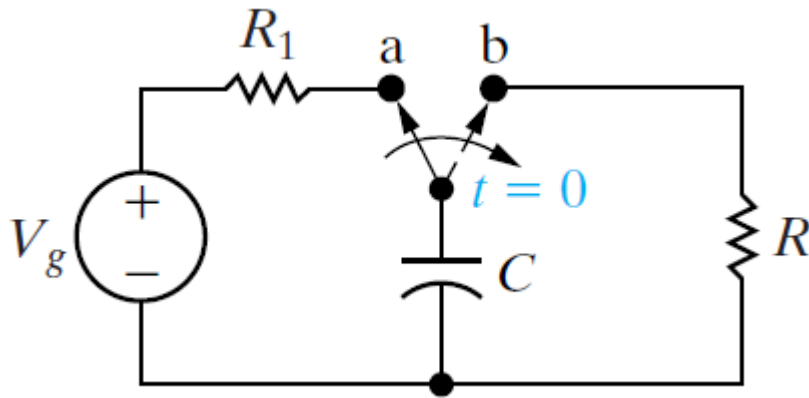
$$i(t) = \frac{V_s}{R} + \left(I_0 - \frac{V_s}{R} \right) e^{-(R/L)t}.$$



RC Circuit Example

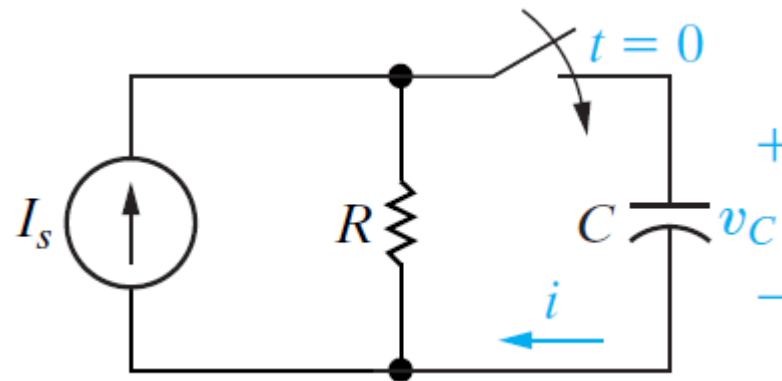
Natural response of RC circuit

Self-reading



$$v(t) = V_0 e^{-t/\tau}, \quad t \geq 0,$$

Step response of RC circuit



$$v_C = I_s R + (V_0 - I_s R) e^{-t/RC}, \quad t \geq 0.$$

End

<https://www.mathtutordvd.com/public/Thevenin-Equivalent-in-Circuit-Analysis.cfm>